Task 2:Linux Boot Process, Port Numbers, and Server Administration Commands

Objective

Booting Process in Linux (Steps in detail)
Important Port Numbers
Some Advanced Server Administrator commands for troubleshooting and monitoring

Part 1: Linux Boot Process

The Linux boot process consists of the following stages, executed in sequence from power-on to a fully usable system:

Step 1 – BIOS/UEFI (Firmware)

When the system is powered on, the motherboard's firmware (BIOS or UEFI) initializes first. It performs POST (Power-On Self-Test), a diagnostic check confirming that essential hardware components (CPU, RAM, keyboard, storage, video) are present and functioning correctly. Once POST completes successfully, the firmware checks its configured boot order and locates a valid boot device.

Step 2 – Bootloader (GRUB2)

The firmware loads the bootloader, GRUB2, from the boot device. GRUB displays a boot menu (if multiple kernel versions are installed), then locates and loads the Linux kernel image (vmlinuz) and the initial RAM filesystem (initramfs) into memory. It passes boot parameters, including the location of the root filesystem, and hands off control to the kernel.

Step 3 – Kernel Initialization

The kernel decompresses and initializes itself, detects hardware, and loads the necessary drivers. It mounts initramfs as a temporary root filesystem, uses it to locate and prepare the real root filesystem (which may involve LVM, RAID, or encrypted storage), and then performs a switchroot to mount the actual filesystem. Finally, the kernel starts the first process, /sbin/init (systemd), which is assigned Process ID 1 (PID 1).

Step 4 – systemd (Init System)

systemd reads its default target (equivalent to the old runlevel concept) and determines all required units (services, mounts, sockets) needed to reach that target. It builds a dependency graph and starts units in parallel wherever possible, rather than sequentially, resulting in faster boot times. systemd continues to supervise all running services throughout the system's

Step 5 – Targets

systemd uses named targets instead of numbered runlevels:

Old Runlevel	Systemd Target	Purpose
0	poweroff.target	Shutdown
1	rescue.target	Single-user/rescue mode
3	multi-user.target	Multi-user, no GUI (typical for servers)
5	graphical.target	Multi-user with GUI
6	reboot.target	Reboot

Step 6 – Login Prompt

Once all required units for the target have started, systemd starts a getty process (or, on GUI systems, a display manager), presenting a login prompt. On a server such as an EC2 instance, this stage makes the system ready to accept SSH connections.

Power On → BIOS/UEFI (POST) → GRUB (loads kernel + initramfs) → Kernel initializes hardware, mounts real root filesystem → systemd (PID 1) starts services → Target reached → Login/SSH ready

Part 2: Important Port Numbers

A port number is a numeric identifier (0–65535) that allows a single machine to run multiple network services simultaneously. While an IP address directs traffic to the correct machine, the port number directs it to the correct application or service running on that machine. The table below lists commonly used port numbers relevant to server administration and networking

Port	Service	Description
20, 21	FTP	File transfer (data and control channels)
22	SSH	Secure encrypted remote access
23	Telnet	Unencrypted remote access (obsolete)
25	SMTP	Sending email between servers
53	DNS	Domain name resolution
67, 68	DHCP	Automatic IP address assignment
80	HTTP	Unencrypted web traffic

110	POP3	Email retrieval
123	NTP	Time synchronization
143	IMAP	Email retrieval with server-side sync
161, 162	SNMP	Network device monitoring
389	LDAP	Centralized directory/authentication
443	HTTPS	Encrypted web traffic
445	SMB	Windows file/printer sharing
3306	MySQL	Database connections
3389	RDP	Windows remote desktop
5432	PostgreSQL	Database connections
6379	Redis	In-memory data store
8080	HTTP-alt	Alternate web/proxy port
27017	MongoDB	Database connections

Ports currently in use on a system can be checked using:

sudo ss –tulnp

Part 3: Advanced Server Administration Commands

These are commonly used Linux commands that help a server administrator monitor system performance, manage running processes, troubleshoot network connectivity, and review system logs. They are grouped below by purpose — resource monitoring, process management, network troubleshooting, log analysis, disk checks, and user/permission checks.

System Resource Monitoring

Commands used to check how much CPU, memory, and disk space a server is currently using, helping identify performance bottlenecks before they cause issues.

- top — Live CPU/memory usage per process
- htop — Interactive, improved version of top
- free -h — Memory usage (human-readable)
- df -h — Disk space usage per filesystem
- du -sh * — Disk usage per file/folder
- uptime — System uptime and load average

Process Management

Commands used to view running processes, check the status of services, stop unresponsive processes, and monitor service-specific logs in real time.

- ps aux — List all running processes
- ps -ef | grep nginx — Find a specific process
- kill -9 <PID> — Force-terminate a process
- systemctl status <service> — Check service status
- journalctl -u nginx -f — Live logs for a specific service

Network Troubleshooting

Commands used to check connectivity, verify DNS resolution, trace the network path to a host, and identify which ports are open or listening on a server.

- ss -tulnp # List listening ports and owning processes
- netstat -tulnp # Older equivalent of ss
- ping <host> # Basic connectivity check
- curl -I <url> # Quick HTTP status/header check
- traceroute <host> # Trace network path to a host
- nslookup <domain> # Basic DNS lookup
- dig <domain> # Detailed DNS lookup

Log Troubleshooting

Commands used to view system and application logs in real time, helping identify the cause of errors or unexpected behavior as they occur.

- tail -f /var/log/messages — Live system log (RHEL)
- tail -f /var/log/nginx/error.log — Live Nginx error log
- journalctl -xe — Recent detailed system logs
- dmesg | tail — Recent kernel/hardware messages

Disk and Filesystem Checks

Commands used to view disk partitions, check which filesystems are currently mounted, and inspect disk layout in detail.

- lsblk — List block devices/partitions
- mount — Show mounted filesystems
- fdisk -l — List disk partitions in detail

User and Permissions

Commands used to check the currently logged-in user, their permissions and group memberships, and recent or active login sessions on the server.

- whoami — Current logged-in user
- id — User's UID/GID and group memberships
- last — Recent login history
- who — Currently logged-in users

Conclusion

This task provided a detailed understanding of the Linux boot process, from firmware initialization through GRUB, the kernel, and systemd, along with key network port numbers and advanced Linux commands used for system monitoring, process management, and troubleshooting. This knowledge builds a practical foundation for diagnosing and resolving real-world server issues.